National College of Ireland

MSc in Cloud Computing, Year 1, MSCCLOUD1_A
MSc in Cloud Computing, Year 1, MSCCLOUD1_B

Release Date: Tuesday 7th October 2025
Submission Deadline: Monday 1st December 2025 at 09:00am

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Cloud Platform Programming
Continuous Assessment (CA): Project

Weight: The assignment will be marked out of 100. This CA represents 60% of the overall marks for this module.

Instructions: This is an individual assessment/project. You are required to submit all the project deliverables through Moodle via the dedicated submission pages.

SUBMISSION DETAILS: All the project deliverables must be submitted via Moodle before the deadline using the dedicated submission pages. **Late submissions will be penalised** unless applied for an extension through NCI360 and approved. The report deliverable should be formed from paragraphs and should NOT contain ordered and/or unordered (e.g., bullet points) lists.

Academic Integrity: IMPORTANT: It is your responsibility to avoid plagiarism. Please read the comprehensive guidelines on academic honesty and academic integrity, and how to avoid plagiarism made available by the NCI Library (https://libguides.ncirl.ie/referencingandavoidingplagiarism).
It is expected that the Submitted Assignment is FULLY THE WORK OF THE STUDENT except where quoted material is clearly indicated. The use of generative Artificial Intelligence (AI) text generators (e.g., ChatGPT, Quillbot, GitHub Copilot, etc.) to generate/rewrite/paraphrase text and/or to generate code could be considered a breach of academic integrity as you are not doing the work yourself and are using AI to carry out the work YOU should be doing. Paraphrasing is a very important part of an assignment as it proves a person understood the information enough to put it into their own words.

NOTE: YOU ARE NOT ALLOWED TO PUBLISH THIS ASSIGNMENT BRIEF OR A PART THEREOF ON ANY WEBSITES. YOU ARE NOT ALLOWED TO PUBLISH/SHARE YOUR SOLUTION WITH OTHERS. All work submitted should be YOUR own. Conferring with others is NOT permitted. This is NOT a collaborative assessment.

TURNITIN: All submissions will be electronically screened for evidence of academic misconduct, e.g., plagiarism, collusion, AI generated text and/or code, misrepresentation, etc. Any submission showing evidence of academic misconduct will be investigated in accordance with the provisions of the Academic Integrity Policy.

Introduction

The assessment of the Cloud Platform Programming module is based on two assessment components:

a) *Project* which represents 60% of the module assessment, and
b) *Terminal Assessment* which represents 40% of the module assessment.

The learning outcomes of the Cloud Platform Programming module are as follows:

LO1. Demonstrate in-depth knowledge of core cloud-based services.
LO2. Critically analyse advantages and disadvantages of different cloud-based technologies/services.
LO3. Formulate and produce new code libraries that implement advanced programming constructs in order to create secure, dynamic, configurable, robust, scalable cloud-based applications.
LO4. Construct and present a complex dynamic cloud-based application through selecting relevant cloud related architectural patterns and services taking into account the evaluation and assessment of application design, development, and testing methodologies.
LO5. Identify and ethically apply best practices for continuous integration, delivery and deployment of cloud-based applications.

This document presents the details of the Project assessment of the Cloud Platform Programming module. The Project assesses LO1, LO2, LO3, LO4, and LO5.

Project Description

For this assignment, assume that a company has commissioned you to develop a complex dynamic cloud-based application in a given domain through selecting relevant cloud related architectural patterns and cloud-based services. Your application must be deployed and hosted on a public Cloud. The given domain is assigned to you based on the penultimate i.e., second to last digit of your Student ID as described in the *Table 1 Industry Sectors.*

Table 1 Industry Sectors

Penultimate (i.e., second to last) digit of Student ID	Industry sector
0	Transportation
1	Electricity, Gas, Steam and Air Conditioning Supply
2	Accommodations and Lodging
3	Retail Trade
4	Electronics
5	Agriculture and Forestry
6	Finance and Economic
7	Insurance
8	Food and Beverage Manufacturing
9	Supply Chain Management

IMPORTANT: Each student must develop a complex dynamic cloud-based application in the correct domain based on the guidelines from *Table 1 Industry Sectors.* **IMPORTANT: This is a submission requirement.** If the incorrect industry sector is chosen, the project will not be valid, and **NO marks** will be provided.

Example: According to Table 1, a student with the student ID = 259876**5**4 would be assigned the industry sector *Agriculture and Forestry* (because the penultimate i.e., second to last digit of that student ID is **5**).

Once you have decided what your application will do, you should go through the following process:

- Define requirements: describe the functional and non-functional requirements of your application.
- Critically analyse and document the architecture of your cloud-based application.
- Implement a cloud-based application that addresses the requirements defined above. This application must employ programmatically **at least five** different cloud services including cloud services for object storage, database, and serverless computing. In addition, the application must use, at least, one new library that you develop in an object-oriented programming language. The new library should provide meaningful functionality to your application.
- Ensure that you implement/code your application to take advantage of suitable cloud architectural patterns.
- Deploy your application to a suitable public cloud platform. The deployed application must not be modified after the submission deadline. The examiner should be able to view your deployed application without having to register for

any account with the public cloud provider where you deployed your application (i.e., the application (**not its source code!)** should be publicly accessible via a URL). This publicly accessible URL should not be shared with anyone else but the examiners of this instance of the Cloud Platform Programming module at NCI.

- You must conduct some independent research and include any relevant bibliography in the accompanying report.

Project Deliverables

You are required to submit all the project deliverables through Moodle via the dedicated submission pages.

You are required to document the process of developing the cloud-based application and reflect on it through the following deliverables:

1. A **project report** in PDF (7-8 pages formatted using the IEEE Conference double-column template[1]) which should include:
    - NCI Project Submission Sheet/ Project Cover Sheet
    - Headline: title of the report, your name, student number, module, programme, and date
    - Abstract – a 150-300-word executive summary of the project and the main results
    - Section 1: Introduction – motivation for your project and its main objectives
    - Section 2: Project requirements: describe the functional and non-functional requirements of your application
    - Section 3: Architectural design aspects of your application – critically analyse and document the architecture of your cloud-based application including the architecture diagram of the application; the architecture diagram should indicate where the different cloud-based services fit into your system/application. Note that the diagram should be created by you based on your own application and cloud services used.
    - Section 4: For each of the cloud-based services used in the application, critically analyse advantages and disadvantages of those cloud-based services and justify the choice of cloud services used.
    - Section 5: Library description – first, present the main purpose of the library you developed. Next, describe the functionalities the library supports in your application. You should include relevant code snippets of where the library is used in your application. Provide the URL to your published library.
    - Section 6: Implementation – document the implementation of the requirements of your application. For each cloud service used in your application, explain the requirement(s)/functionality/functionalities that particular cloud service supports, and include code snippets to show how you integrated that cloud service in your application.
    - Section 7: Continuous integration, delivery, and deployment of your application
        - Include in the report the URL to your deployed application
        - Note that at all times you **must use a private repository** for versioning control (e.g., GitHub, GitLab, etc.)
    - Section 8: Conclusions including findings/interpretations – what did you learn and find out? Include a short reflection on developing this project. If you were to implement this project again, what would you do differently?
    - Section 9: References – a complete list of academic works and/or online materials used in the project. References should be included as in-text citations using the IEEE referencing style. Note that a good starting point to find academic works is the NCI Library Guide on Cloud Computing at https://libguides.ncirl.ie/cloudcomputing

IMPORTANT: Anything after Page 8 will not be considered i.e., it WILL NOT BE MARKED! (Note that the NCI Project Submission Sheet/ Project Cover Sheet is not considered as part of the page count.)

[1] https://www.ieee.org/conferences/publishing/templates.html

2. The **source code artefacts** submission (a ZIP file) should include:
    - Source code of the application (including comments)
      Note: Please include substantial meaningful comments in YOUR source code to document your ORIGINAL contributions
    - Source code of the library (including comments)
      Note: Please include substantial meaningful comments in YOUR source code to document your ORIGINAL contributions

3. **In-person project presentation and demonstration, to be held in class during the submission week as per the CA schedule. It should include the following:**
    - Project presentation slide deck
    - Demonstration of your Cloud Platform Programming project
    - As part of the project presentation and demonstration you will be required to answer questions

    Notes:
    1. **You are required to submit the project presentation slide deck together with all the other deliverables by the submission deadline**
    2. **You must present/demonstrate your project in-person to be marked 'Present' for this project. Those who do not present/demonstrate their project in-person will be marked as *'Absent without permission'* and the other project deliverables will NOT BE GRADED.**

Assessment Criteria

The Project will be assessed based on the assessment criteria shown in Table 2 and the marking rubric shown in Table 3.

Table 2 Assessment Criteria

Architectural Design	10%	Critique possible architectures for the cloud application and discuss the chosen architecture.
Cloud Services	15%	Evaluate, select, and use in your project multiple cloud services.
Library Creation	15%	Design and implement at least one new library documenting the usefulness of the library in the context of your application.
Implementation	20%	Develop a complex dynamic cloud-based application in which at least five cloud services are integrated programmatically to support the features of the application.
Deployment	10%	Deploy your application to a suitable public cloud platform.
Conclusions and Findings	5%	The report should incorporate conclusions including findings/interpretations.
Project Demonstration & Answers to Questions	25%	Present the cloud-based application. Demonstrate the cloud-based application you developed, presenting the dynamic characteristics of it and highlighting the cloud-based services used to support the functionalities of your application. Answer the questions received from the lecturer.

Table 3 Marking Rubric

Grade Criterion	H1 (> 70%)	60-69%	50-59%	40-49%	Fail (< 40%)
Architectural Design: 10%	Architectural considerations have been well prepared, explored and critically analysed.	Architectural considerations have been prepared, explored and analysed.	Architectural considerations have been prepared and mostly explored, but not analysed.	Architectural considerations have been prepared but somewhat trivial.	Architectural considerations are not evident.
Cloud Services: 15%	Excellent/very good critical analysis of relevant cloud services. Evidence of breadth and depth in the reviewed cloud services. Excellent justification/discussion for/on the choice of cloud services used.	Good critical analysis of relevant cloud services. Evidence of breadth in the reviewed cloud services. Good justification/discussion for/on the choice of cloud services used.	Adequate critical analysis of relevant cloud services. Reasonable use of concepts, but there is a lack of breadth in the reviewed cloud services. Adequate justification/discussion for/on the choice of cloud services used.	Limited critical analysis of cloud services. Limited use of concepts. Limited justification/discussion for/on the choice of cloud services used.	Very limited and poor or inexistent critical analysis of cloud services. Very limited use of concepts. Very limited/ poor or inexistent justification/discussion for/on the choice of cloud services used.
Library Creation: 15%	Excellent/very good identification and development of a library in terms of providing suitable/meaningful functionalities to the application. Excellent/very good documentation/description of the library.	Good identification and development of a library in terms of providing suitable/meaningful functionalities to the application. Good documentation/description of the library.	Adequate identification and development of a library in terms of providing suitable/meaningful functionalities to the application. Adequate documentation/description of the library.	Weak identification and development of a library in terms of providing suitable/meaningful functionalities to the application. Weak documentation/description of the library.	Poor identification and development of a library in terms of providing suitable/meaningful functionalities to the application. Poor documentation/description of the library.
Implementation: 20%	Excellent/very good implementation of a complex cloud-based dynamic application in terms of appropriate methodology and cloud-based services used programmatically to support the features of the application. Excellent/very good discussion and documentation of the implementation of the application.	Good implementation of a complex cloud-based dynamic application in terms of appropriate methodology and cloud-based services used programmatically to support the features of the application. Good discussion and documentation of the implementation of the application.	Adequate implementation of a complex cloud-based dynamic application in terms of appropriate methodology and cloud-based services used programmatically to support the features of the application. Adequate discussion and documentation of the implementation of the application.	Weak implementation of a complex cloud-based dynamic application in terms of appropriate methodology and cloud-based services used programmatically to support the features of the application. Weak discussion and documentation of the implementation of the application.	Poor implementation of a cloud-based dynamic application in terms of appropriate methodology and cloud-based services used programmatically to support the features of the application. Poor discussion and documentation of the implementation of the application.
Deployment: 10%	The application is deployed on a suitable public cloud platform. All the functionalities of the application are working properly on the deployed application and are available via the deployed URL.	The application is deployed on a suitable public cloud platform. Some of the functionalities of the application are working on the deployed URL.	Efforts have been made to deploy the application on a suitable cloud platform, but it is not working properly as required.	Efforts have been made to deploy the application on a suitable cloud platform, but this may be unsuccessful.	No efforts have been made to deploy the application on a cloud platform.
Conclusions and Findings: 5%	Excellent/very good conclusions, insightful findings and reflection.	Good conclusions and good discussion of findings and reflection.	Adequate conclusions and adequate discussion of findings and reflection.	Limited/weak conclusions and limited/weak discussion of findings and reflection.	Very limited and poor/inexistent conclusions. Very limited and poor/inexistent discussion of findings and reflection.
Project Demonstration & Answers to Questions: 25%	Excellent well directed presentation and demonstration with impeccable handling of questions.	Clear presentation and demonstration with good handling of questions.	Neat oral presentation and demonstration and acceptable handling of questions.	Poor oral presentation and demonstration and weak handling of questions.	Unacceptable oral presentation and demonstration and poor handling of questions.